THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number written in base 6 ending in the digit 0 is always divisible by: 1

(a) 5 **(b)** 7 **(c)** 10 **(d)** 6

2. If $(56)_b = 46$ in decimal (a two-digit number in base b), then b equals: 1

(a) 9 **(b)** 7 **(c)** 8 **(d)** 10

3. Weights of 1, 2, 4, 8, ... kg (powers of 2) are available. The least number of weights needed to weigh any whole number of kg up to 200 kg is: 1

(a) 8 **(b)** 7 **(c)** 20 **(d)** 9

4. Which of the following numbers is the smallest? 1

(a) 13_5 **(b)** 11_8 **(c)** 10_{10} **(d)** 1011_2

5. The number of 1s in the binary form of 341 is: 1

(a) 6 **(b)** 7 **(c)** 5 **(d)** 4

6. The Roman numeral for 1649 is: 1

(a) MDCXLIX **(b)** MDCXLVIII **(c)** MDCL **(d)** MDCLIX

7. The number that comes immediately after 110011_2 in the binary system is: 1

(a) 110100_2 **(b)** 110101_2 **(c)** 110010_2 **(d)** 1100110_2

8. A number written in base 8 ending in the digit 0 is always divisible by: 1

(a) 7 **(b)** 10 **(c)** 9 **(d)** 8

9. **Assertion (A):** The number 13 is written 1101_2 in binary. 1

Reason (R): Repeated division of 13 by 2 gives remainders 1, 0, 1, 1 from the first division, and these read upwards form the binary number.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $111_2 = 7$. 1

Reason (R): The binary system uses only the digits 0 and 1.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Convert the Roman numeral CLXIX to binary. 2

12. Find the base b if $(61)_b = 55$ in decimal. 2

13. Write 118 in binary and count the number of 1s in it. 2

14. Convert the Roman numeral CCCXXX to binary. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 111 g? Explain using binary numbers. 3

16. Add 1012_4 and 311_4 (numbers in base 4) and give the answer in base 4. 3

17. How many different numbers can be written using at most 4 digits in base 6? What is the largest of them in decimal? 3

OR

A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 58 g? Explain using binary numbers. 3

18. Add 303_7 and 201_7 (numbers in base 7) and give the answer in base 7. 3

OR

How many different numbers can be written using at most 3 digits in base 8? What is the largest of them in decimal? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher writes the number 2143_5 on the board, in base 5, and asks students to analyse it. 5

(i) Which digits are allowed in base 5?

(ii) Write 2143_5 in expanded form.

(iii) Find its value in decimal.

(iv) Express the decimal number 59 in base 5.

20. In some base b the number $(321)_b$ equals 57 in decimal. Find b and verify your answer by expanding. 5

OR

Subtract $2112_4 - 1013_4$ in base 4. Verify your answer in decimal. 5

END OF QUESTION PAPER — **Total: 40 Marks**